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Grade	**0** out of 100

Question **1**

Not answered

Marked out of 1

A psychologist designs a new questionnaire to detect Asperger's syndrome in school going boys. The extent to which this psychometric questionnaire measures what it was originally destined to measure is called

Select one:

- Specificity
- Reliability
- Sensitivity
- Validity
- True positive rate

Your answer is incorrect.

Validity is the extent to which a test measures what it claims to measure. It is very crucial for any test used in clinical and research practice to be valid iso that the results can be accurately used and interpreted

The correct answer is: Validity

Question 2

Not answered

Marked out of 1

A radiographer devises a new method to test for the presence of white matter hyperintensities in brain images of bipolar patients. Which of the following qualities of the new test will depend upon the precision of the measurement

Select one:

- Validity
- Systematic error
- Test duration
- Reliability
- Cost of the test

Your answer is incorrect.

The more precise the results are, the more reliable and repeatable the test will be. A precise result can still be biased (with systematic error). Precision does not relate to cost or duration directly.

The correct answer is: Reliability

Question 3

Not answered

Marked out of 1

Validity and reliability are two important properties when making inferences from a study. Which of the following is a correct statement

Select one:

- Precision is a sufficient condition for validity
- Reliability is a necessary condition for validity
- A reliable test cannot be valid
- Ability to measure consistently is also called accuracy
- An invalid test is always unreliable

Your answer is incorrect.

We can have a valid but unreliable test; we can have an invalid but reliable test too. But in most cases a reliable test must have at least some degree of validity though the relationship is not all-or-none. Ability to measure consistently is called reliability.

The correct answer is: Reliability is a necessary condition for validity

Question **4**

Not answered

Marked out of 1

In a case-control study the examining association of reserpine and breast cancer, controls with cardiovascular disease were excluded but this criterion was not applied to cases. The study concluded an association between reserpine and breast cancer. Which of the following is primarily affected due to the study design?

Select one:

- Predicitive validity
- Reliability
- Internal validity
- Generalisability
- External validity

Your answer is incorrect.

A study's internal validity is the accuracy of its conclusions about an intervention's effects on a given group of subjects under a study's specific circumstances. It depends on both a trial's design and conduct. For a study to have internal validity, the difference in the interventions should be the sole source of the outcome differences. A study may lack internal validity if the differing outcomes are due to factors other than the difference in the interventions. Such factors might be unequal distributions of prognostically important subject characteristics, differences in intervention delivery, or inconsistent measurement of significant elements such as comorbidities or outcomes. This type of lack of control creates a bias that undermines a clinician's ability to trust that what the study says is true. http://www.medscape.com/viewarticle/579200_2

The correct answer is: Internal validity

Question 5

Not answered

Marked out of 1

Which of the following situations will not be affected by the phenomenon of regression to mean when a repeat measurement and comparison is undertaken?

Select one:

- Crime counts before and after in two areas with differential crime rates where one with the highest rates gets a new Crime Reduction Intervention Team.
- Choosing patients for an open trial of antihypertensives based on a threshold for systolic blood pressure
- Prescribing a new lipid lowering drug for primary care patients in a practice based on highest cholesterol measurements in the preceding month.
- Choosing subjects for a special tuition based on the lowest mathematical ability as measured by a test performance
- An RCT for effect of topical steroids on self injury related skin excoriations

Your answer is incorrect.

If a randomly occurring parameter is measured to select a subgroup of extreme values to repeat the measurement, the second measurement will be closer to original mean and not in the extreme. This is known as regression to mean. It applies to most continuously distributed biological parameters such as BP, heart rate, etc. Consider dice throwing. If you select a group of 10 people who threw the highest value (6 in a dice) from a group of 100, and ask them to re-throw the dice, this time, their mean will be lower than 6. Similarly, if all 'one' throwers were grouped and asked to re-throw their group mean will regress towards the true mean (i.e. higher than 1). The simplest way of eliminating this is comparing with a control group chosen randomly - both groups will suffer regression to mean, but as they are equivalent on the baseline, the differences will even out at a comparison. See BMJ 2009; 338:b1164

The correct answer is: An RCT for effect of topical steroids on self injury related skin excoriations

Question 6

Not answered

Marked out of 1

Which of the following is considered as a threat to reliability than the validity of study results?

Select one:

- Confounding
- Regression to mean
- Random error
- Systematic error
- Selection bias

Your answer is incorrect.

External validity or reliability deals with the applicability of a study's conclusions to the real world, beyond the conditions imposed by the trial. This is the generalizability of research and is affected by the random error (imprecision) of study measures.

The correct answer is: Random error

Question 7

Not answered

Marked out of 1

3rd-grade children with the worst reading skills are selected to participate in a nationwide reading skills program (NRSP). At the end of the program, it is concluded that positive changes take place when extra-educational efforts incorporated in the NRSP are used. Which of the following methods could have yielded more accurate results?

Select one:

- Recruiting 4th grade children
- Recruiting 2nd grade children
- Testing the selected children again before the start of the program to obtain a baseline
- Selecting the highest performing children in 3rd grade
- Using p values instead of confidence limits when reporting outcome

Your answer is incorrect.

Had the children been tested again before the start of the program, they might have obtained less extreme scores, even after ruling out repeated testing effects. This is called regression towards the mean. It is an important threat to the validity of a study (http://www.wikidoc.org/index.php/Internal_validity).

The correct answer is: Testing the selected children again before the start of the program to obtain a baseline

Question 8

Not answered

Marked out of 1

If a psychometric test is proposed to measure a single construct which of the following is true?

Select one:

- The items should not correlate with the total score
- The test should not correlate with other measures of the same construct
- The score must predict future performance
- There must be a reliable alternative form
- The items should correlate with the total score.

Your answer is incorrect.

Correlating with other tests that examine same construct is concurrent validity. It is desirable. If items correlate with one another, good construct validity may be present. Though predictive validity is desirable, it is not a neceßity for a good test.

The correct answer is: The items should correlate with the total score.

Question 9

Not answered

Marked out of 1

A psychology research team develops a test of creativity. Which of the following represent convergent and discriminant validity respectively

Select one:

- The test correlates highly with another test of creativity and is uncorrelated with verbal intelligence
- The test correlates highly with verbal intelligence test but not with another test of creativity
- The test has item scores correlating with the total score of creativity but items measure different constructs related to creativity
- The test correlates highly with another test of creativity and also with verbal intelligence
- The test does not correlate highly with another test of creativity and is uncorrelated with verbal intelligence

Your answer is incorrect.

Convergent and discriminant validity are both considered subcategories or subtypes of construct validity. The important thing to recognize is that they work together -- if you can demonstrate that you have evidence for both convergent and discriminant validity, then you've by definition demonstrated that you have evidence for construct validity. But, neither one alone is sufficient for establishing construct validity (Retrieved from http://www.socialresearchmethods.net/kb/convdisc.php). "To establish convergent validity, you need to show that measures that should be related are in reality related. To establish discriminant validity, you need to show that measures that should not be related are in reality not related". http://www.socialresearchmethods.net/kb/convdisc.php

The correct answer is: The test correlates highly with another test of creativity and is uncorrelated with verbal intelligence

Question **10**

Not answered

Marked out of 1

If MRCPsych Paper B accurately indicates participants' scores on MRCPsych CASC exams taken at a later date, then MRCPsych Paper B is said to have which of the following

Select one:

- Concurrent validity
- Retrospective validity
- Content validity
- Predictive validity
- Face validity

Your answer is incorrect.

In predictive validity, we assess the ability of a measurement scale to predict something it should theoretically be able to predict. For instance, we might theorize that a measure of math ability should be able to predict how well a person will do in an engineering-based profession. We could give our measure to experienced engineers and see if there is a high correlation between scores on the measure and their salaries as engineers. A high correlation would provide evidence for predictive validity -- it would show that our measure can correctly predict something that we theoretically think it should be able to predict. http://www.socialresearchmethods.net/kb/measval.php

The correct answer is: Predictive validity

Question **11**

Not answered

Marked out of 1

Which of the following statements accurately describes test-retest reliability?

Select one:
- Measure of consistency of scores obtained from two equivalent halves of the same test
- Measure of consistency with which a test measures a single construct or concept
- Measure of degree of agreement between two or more scorers, judges, or raters
- Measures of correlation among various items of the test
- Measure of consistency of test scores over time.

Your answer is incorrect.

We estimate test-retest reliability when we administer the same test to the same sample on two different occasions. This approach assumes that there is no substantial change in the construct being measured between the two occasions. The time gap between two measurements is crucial - longer the gap, poorer the reliability.

The correct answer is: Measure of consistency of test scores over time.

Question **12**

Not answered

Marked out of 1

What does Cronbach's alpha measure?

Select one:

- Internal consistency
- Construct validity.
- Criterion validity.
- Inter-rater reliability.
- Test-retest reliability.

Your answer is incorrect.

Cronbach's alpha measures internal consistency (a type of reliability) - it can take any value between -infinity and +1.

The correct answer is: Internal consistency

Question **13**

Not answered

Marked out of 1

A depression rating scale A gives a score of 10, 12, 14, 11 and 12 when used in the same patient by different raters. A different scale B gave values of 28, 7, 6, 29, and 16. When a gold standard test is used, the depression score was equivalent to 12. Choose one right statement from the below:

Select one:

- Scale B is accurate but not precise
- Scale A is both precise and accurate
- Scale A and B are both accurate but not precise
- Scale A is very precise but not accurate
- Scale B is both precise and accurate

Your answer is incorrect.

Precision is the degree to which a calculated central value (e.g. mean) varies with repeated sampling. The narrower the variation, the more precise the value is said to be. Factors reducing precision includes 1. Having wider limits of the interval (range) 2. Expecting higher confidence interval (e.g. 99.7% versus 95%). Accuracy refers to the correctness of the mean value i.e. how close it is to the true population value. Precision is comparable to reliability while accuracy is comparable to validity. Ref: http://www.engr.mun.ca/~cdaley/1000/measurem.htm

The correct answer is: Scale A is both precise and accurate

Question **14**

Not answered

Marked out of 1

A psychiatrist wants to examine the validity of MRCPsych theory exams that purportedly test the knowledge base of a psychiatric trainee. She chooses to examine the degree of association between paper A scores and paper B scores. Which of the following is she testing?

Select one:

- Content validity
- Divergent validity
- Face validity
- Convergent validity
- Divergent validity

Your answer is incorrect.

Convergent validity is a part of construct validity. Two measures of the same construct should be, in theory, related to each other. We can test this correspondence or convergence using correlation metrics (Note that the construct examined by the two measures is the same - theoretical knowledge base). Predictive validity refers to the ability of a test to predict future group differences, according to current group differences in score. E.g., can paper A score predict ARCP outcome or CASC scores (note that the two constructs examined here are not the same, they may be related).

The correct answer is: Convergent validity